



Course Name: Computer Networks and Internet Protocol

Assignment 3 - Week 3 (Jan 2025)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1

What kind of sequence number does TCP use?

- a) byte-oriented sequence number
- b) packet-oriented sequence number
- c) Randomly generated fixed sequence numbers
- d) none of them

Answer: (a)

Detailed solution:

TCP uses byte-oriented sequence numbering to ensure reliable, ordered, and efficient data transmission over a network.

QUESTION 2

Identify the FALSE statement(s) in the following-

- I. The transport layer functionalities (e.g., connection establishment, flow control, error detection) operate at an abstract level using data structures, algorithms, and system calls implemented in the OS kernel.
- II. The operating system (OS) kernel does not play any role in supporting the transport layer functionalities of the TCP/IP protocol stack.

- a) Only (I)
- b) Only (II)
- c) Both (I) and (II)
- d) Neither (I) nor (II)

Answer: (b)

Detailed solution:

The transport layer relies on data structures (e.g., buffers, sequence number tables) to manage connections, algorithms (e.g., sliding window for flow control, checksum for error detection) to ensure reliable communication, and system calls (e.g., socket(), send(), recv()) to interact with the OS kernel, which implements these functionalities abstractly for application-level use.

QUESTION 3

Which of the following services is NOT supported by the transport layer?

- a) End-to-end packet delivery
- b) Ordered packet delivery
- c) Reliable data delivery
- d) Forwarding the datagram from one hop to another hop in the network

Answer: (d)

Detailed solution:

The transport layer provides services like End-to-end packet delivery, Ordered packet delivery, Reliable data delivery, Connection establishment, and Flow & Congestion control etc. The hop-to-hop network packet forwarding is supported by data link layer and network layer.

QUESTION 4

What can be a primary mechanism used to ensure the reliable data transfer in a stop-and-wait algorithm (Noisy Channel)?

- a) Employing only error detection and correction codes without requiring acknowledgements.
- b) Increasing the timeout interval.
- c) Using a sequence number to identify each packet uniquely and waiting for an acknowledgement for each packet before sending the next packet.
- d) Using a sliding window to send multiple packets without waiting for their acknowledgements.

Answer: (c)

Detailed solution:

Stop-and-Wait (Noisy Channel) flow control algorithm uses a sequence number to individually identify each frame and corresponding acknowledgement. It ensures reliability by attaching a **sequence number** to each packet and transmits a packet & waits for an **acknowledgment (ACK)** from the receiver before sending the next packet. If the ACK is not received within a timeout period, the sender retransmits the packet.

QUESTION 5

A data of 40 bytes need to be delivered using TCP protocol but a sender can send a segment of maximum size 12 bytes only. Identify the sequence number of the last segment formed if the sequence number field uses 5 bits only? [Assume sequence number starts from 0]

- a) 1
- b) 4
- c) 12
- d) 31

Answer: (b)

Detailed solution:

TCP uses byte sequencing. It treats data as a continuous stream of bytes rather than discrete packets. Since the sequence number field uses 5 bits only, it produces sequence numbers ranging from 0 to 31.

The first segment of 12 bytes will use the sequence number range (0-11),

Next segment of 12 bytes will use the sequence number range (12-23),

Next segment of 12 bytes will use the sequence number range (24-31) & (0-3), and

The last segment of 4 bytes will use the sequence number range (4-7).

The sequence number in a TCP segment header refers to the first byte of data in that segment relative to the start of the stream.

QUESTION 6

A sender uses a sliding window protocol with a window size of 5. The range of sequence numbers follows modulo 8 (the sequence number starts from 0 to 7). If the receiver acknowledges packet 1, what will be the sender's new window range?

- a) 1-5
- b) 2-6
- c) 3-7
- d) 0-4

Answer: (b)

Detailed solution:

When the receiver acknowledges the packet, resulting in a forward slide of the sender's window. The sliding window now starts at 2 (the next unacknowledged sequence number) till the allowed window size.

QUESTION 7

Determine whether the following information is True or False?

“During the three-way handshaking of connection establishment procedure (from Host 1 to Host 2), Delayed duplicate SYN can be handled by TCP at Host 2 simply by ignoring it, as the sequence number is invalid”

- a) True
- b) False

Answer: (a)

Detailed solution:

The TCP 3-Way Handshake is a fundamental process that establishes a reliable connection between two devices over a TCP/IP network. During the handshake, the client and server exchange initial sequence numbers. It uses sequence numbers to identify and validate the legitimacy of connection requests. A delayed duplicate SYN will have an outdated sequence number that doesn't match the current connection state at Host 2 and can be ignored as invalid sequence number.

QUESTION 8

TCP instance uses a sliding window protocol in the noisy channel. A timeout occurs due to the loss of a segment (or the acknowledgement). Mark the RIGHT statement among the following-

- a) In the case of Go-Back-N ARQ, if any segment of the sliding window is lost, all the segments of the sliding window are retransmitted while in Selective Repeat ARQ, only the lost packets are selectively transmitted.
- b) In the case of Go-Back-N ARQ, if any segment of the sliding window is lost, only the lost packets of the sliding window are selectively retransmitted while in Selective Repeat ARQ, all packets of the sliding window are retransmitted.
- c) In both cases, Go-Back-N ARQ and Selective Repeat ARQ, all packets of the sliding window are retransmitted.
- d) In the both cases, Go-Back-N ARQ and Selective Repeat ARQ, only lost packets of the sliding window are selectively retransmitted.

Answer: (a)

Detailed solution:

In the case of Go-Back-N ARQ, if any segment of the sliding window is lost, all the segments of the corresponding sliding window are retransmitted while in Selective Repeat ARQ, only the lost packets are selectively transmitted.

QUESTION 9

Consider a sender using the Go-Back-N ARQ protocol with a window size 5. The sequence number ranges from 0 to 7 (modulo 8). The sender initially transmits packets with the sequence numbers- 0, 1, 2, 3, and 4. During the transmission, packet 3 is lost in the network. The sender receives a cumulative ACK for the packet 2. The sender continues sending packet 5, 6, and 7 but the receiver denies accepting them due to the loss of packet 3.

Identify the range of packets that will be retransmitted by the sender after the timeout of packet 3?

- a) 0, 1, 2, 3, and 4.
- b) 2, 3, 4, 5, and 6.
- c) 3, 4, 5, 6, and 7.
- d) 1, 2, 3, 4, and 5.

Answer: (c)

Detailed solution:

In Go-Back-N ARQ, when a timeout occurs, the sender retransmits all packets starting from the lost packet to all permissible packets in accordance with the window size.

QUESTION 10

A client and a server establish a connection using TCP's three-way handshake. The following details are given:

- Initial sequence number of the client: 100 and initial sequence number of the server: 300.
- RTT (Round Trip Time): 80ms.
- Each TCP segment carries 40 bytes of header information.
- The server sends a SYN-ACK segment, and the acknowledgement (ACK) segment from the client is delayed due to congestion, taking 100ms to reach the server.

How much time does it take to establish the connection under such a scenario?

- a) 100ms
- b) 280ms
- c) 180ms
- d) 260ms

Answer: (c)

Detailed solution:

In three way handshaking, the connection is established after three steps. In first step, sender sends a connection request SYN packet to the server. Server responds to client with SYN+ACK packet. Finally, sender sends a ACK packet to the server. Since $RTT = 80ms$ so for one way packet transmission will take $40ms$. Since, there is a delay during transmission of ACK packet from sender to server due to congestion and it takes greater time i.e. $100ms$. So, the total time needed to complete this connection procedure will be (time taken for SYN packet + time taken for SYN+ACK packet + time taken for ACK packet) i.e. $(80/2ms + 80/2ms + 100ms = 180ms)$.
